4. Projectile Motion

And working out a safety distance you can defend

Turning a bang into a number

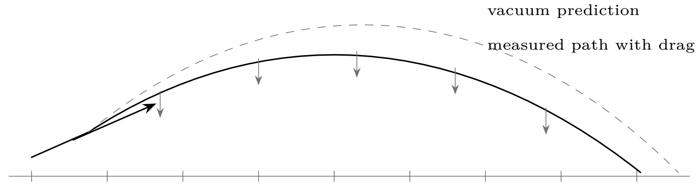
Lesson 4 · about 90 minutes · bench work, then range work with the adult

What you need

- A long tape measure or a measuring wheel
- A protractor or a phone inclinometer app
- A stopwatch, and a phone that shoots slow motion
- Marker flags or cones
- A notebook — this lesson is mostly recording

Predict first

At what launch angle does a projectile go furthest? Most people say 45 degrees. Write down whether you think that is exactly right for a potato, and if not, whether the best angle is higher or lower — and why.



Do it — on the bench first

1. **Drop and time.** Drop a ball from a measured height and time the fall. Repeat five times and average. Compute g from $h = \frac{1}{2}gt^2$. You should land near $9.8\,\mathrm{m/s^2}$.
2. **Horizontal launch.** Roll a ball off a table at various speeds. It always takes the *same* time to hit the floor. Horizontal and vertical motion are independent, and that single fact is the whole of projectile motion.
3. **Angle trial.** With a rubber-band launcher or a hose, fire at 20, 30, 40, 45, 50, 60 and 70 degrees and measure each range. Plot range against angle.

Then on the range

1. **Fire flat, from a measured height.** Muzzle horizontal, height h measured. Time the fall from h, then measure how far downrange it lands. Muzzle velocity is simply distance divided by that fall time — no clever equipment required.
2. **Or film it.** Slow motion at 240 frames per second, with two marker flags a known distance apart in shot. Count frames between them. This is the more accurate method and boys will prefer it.
3. **Angle series.** Three shots each at 30, 40, 45 and 50 degrees, same fuel dose, and record every range. Plot it.
4. **Compare with Lesson 3.** You predicted a muzzle velocity from the gas law. How close was it? If it is much lower, where did the energy go?

What it means

Two independent motions	Horizontal at constant speed, vertical accelerating downward. They do not affect one another, and every projectile problem is those two treated separately and then combined.
Range in a vacuum	$R = v^2 \sin(2\theta)/g$, which peaks at exactly 45 degrees. Notice the v^2: doubling muzzle velocity *quadruples* range. Small increases in fuel efficiency matter a great deal downrange.
Range in air	Not 45 degrees. Drag takes energy out for the whole flight, so a flatter, shorter flight loses less — the real optimum for a blunt, light projectile is usually somewhere between 35 and 42 degrees. Your measurements should show this, and it is a genuinely satisfying disagreement with the textbook.
A potato is a bad projectile	Light for its size, blunt, irregular and rough. Drag dominates, so it decelerates hard and falls well short of the vacuum prediction. Compare your measured range with v^2/g and see how much air took.
The energy	$\frac{1}{2}mv^2$. A 200 g potato at 100 m/s carries about 1000 J — comparable to a handgun round, with far more mass and less speed. This is the number that justifies every rule on the **Safety** sheet.
Your safety distance	Fire at 45 degrees and measure the longest range you can produce. Then **double it** and treat that as the minimum clear ground. A projectile does not care what you intended.

Break it

Change the projectile	A potato, a wet rag wad, a tennis ball, a carrot. Same launcher, same fuel, and very different ranges. Which variables changed — mass, drag, seal — and which mattered most?
Fuel versus range	Overlay your Lesson 2 fuel-dose series on the range plot. The peak should be in the same place.
Predict, then check	Before the last shot of the day, predict the range from your own numbers and mark the spot with a flag. Then fire.
Wind	Fire the same shot with and against a steady wind. A light, blunt projectile is affected far more than they expect.

The point of the numbers

The reason to measure all this is not the physics grade. It is that once you have measured a muzzle velocity, computed the energy and seen where a 45-degree shot actually lands, the safety distance stops being a rule somebody imposed and becomes a number you worked out yourself. Nobody argues with their own arithmetic.